BeginPay for WooCommerce

Catalyst Fund 12 - Milestone 3 Report

WooCommerce Cardano Payment Gateway

Executive Summary

This report documents the successful implementation of BeginPay, a modern Cardano (ADA) payment gateway plugin for WooCommerce, fulfilling the requirements of Catalyst Fund 12 Milestone 3. The plugin enables any WooCommerce merchant to accept ADA payments with CIP-30 wallet integration, real-time fiat-to-ADA conversion, QR code fallback, and on-chain payment verification via Blockfrost.

3	11	~10KB	v2.3
Payment Methods	Wallets Supported	TX Builder Size	Plugin Version

Implemented Features

1. CIP-30 Wallet Connection

Full CIP-30 dApp connector integration supporting 11 Cardano wallet extensions including Nami, Eternl, Lace, Flint,

2. In-Browser Transaction Building

Lightweight CBOR transaction builder (~10KB) that constructs valid Cardano Babbage-era transactions directly in the

Typhon, GeroWallet, Yoroi, Vespr, Begin, and NuFi. Automatic wallet detection, one-click connection, and visual wallet status display.

● Complete

browser. Replaces the need for the 27MB Cardano Serialization Library while producing byte-identical output. Handles UTXO selection, fee estimation, change output, and TTL calculation.

● Complete

3. Real-Time ADA/Fiat Conversion

Live exchange rate fetching from CoinMarketCap API with support for all WooCommerce currencies. Cart total is automatically converted to ADA with a one-click rate refresh button. Exchange rate and ADA amount are persisted to order metadata for accounting.

● Complete

4. QR Code & Copy Address Fallback

Three-tab payment interface offering Connect Wallet, QR Code scan, and Copy Address methods. Copy-to-clipboard for both merchant address and ADA amount with visual feedback. QR code configurable via admin settings.

● Complete

5. On-Chain Payment Verification

Blockfrost API integration for real-time on-chain transaction verification. Automatic polling checks transaction status, confirmation count, and verifies the correct ADA amount was sent to the merchant address. Order status auto-updates from "on-hold" to "processing" upon confirmation.

● Complete

6. HPOS-Compatible Order Management

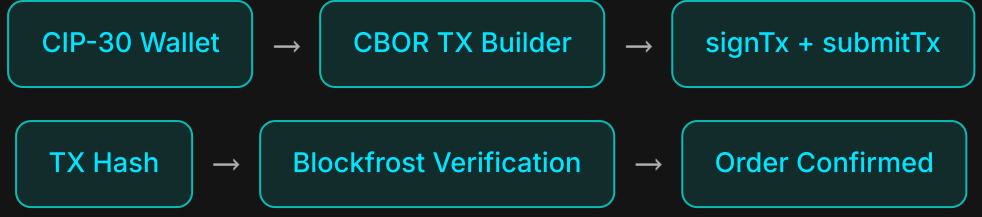
Full compatibility with WooCommerce High-Performance Order Storage. All order metadata (TX ID, ADA amount, exchange rate, payment method, verification status) stored via HPOS-safe methods. Admin order view, thank-you page, and email notifications include Cardano payment details.

● Complete

Technical Architecture

Payment Flow

Checkout Page → CoinMarketCap API → ADA Conversion



Core Components

```
beginpay__func.php - Plugin bootstrap, script enqueuing, localization, and API data fetching
inc/class-wc-payment-gateway-beginpay.php - WC_Payment_Gateway extension with admin settings, validation, and payment processing
inc/beginpay-checkout-description.php - Modern checkout UI template with three-tab payment interface
inc/beginpay-ajax.php - AJAX endpoints for rate refresh, Blockfrost verification, and order confirmation
inc/beginpay-invoice-details.php - HPOS-compatible order meta display for admin, thank-you page, and emails
js/beginpay-cbor.js - Lightweight CBOR encoder/decoder and Cardano transaction builder (~10KB)
js/beginpay-wallet.js - CIP-30 wallet detection, connection, TX signing, polling, and UI management
css/custom.css - Modern responsive styling with card-based layout and animated status indicators
```

CBOR Transaction Builder

A custom lightweight CBOR encoder that builds valid Cardano Babbage-era transactions without external dependencies. The output is byte-for-byte identical to transactions produced by the official Cardano Serialization Library (CSL v11.5.0).

> 1. Retrieve UTXOs from the connected wallet via CIP-30 `getUtxos()`
> 2. Parse each UTXO's CBOR to extract transaction hash, index, and lovelace amount
> 3. Select UTXOs using a greedy algorithm (largest-first) until payment + fee + min-UTXO is covered
> 4. Decode the merchant's bech32 address to raw hex bytes using an inline bech32 decoder
> 5. Build the transaction body as a CBOR map: inputs (key 0), outputs (key 1), fee (key 2), TTL (key 3)
> 6. Construct the full Transaction: `[body, empty_witness_set, is_valid=true, null_auxiliary]`
> 7. Call `signTx(cbor_hex, partialSign=true)` on the wallet API

8. Handle both full-transaction and witness-set-only responses from `signTx`
9. Submit the signed transaction via `submitTx()` and capture the TX hash

Blockfrost Verification Flow

1. After TX submission (or manual TX ID entry), polling begins every 10 seconds
2. AJAX endpoint calls Blockfrost `GET /txs/{hash}` to check transaction existence
3. If found, fetches `GET /blocks/latest` to calculate confirmation count
4. Calls `GET /txs/{hash}/utxos` to verify the merchant address received the correct ADA amount
5. Once confirmed (≥1 confirmation), the WooCommerce order status transitions from "on-hold" to "processing"
6. TX ID, ADA amount, exchange rate, and confirmation count are saved to order metadata
7. Checkout form auto-submits after on-chain confirmation, redirecting to the thank-you page

Supported Wallets

Wallet	CIP-30 Key	Status
Nami	`nami`	● Supported
Eternl	`eternl`	● Supported
Lace	`lace`	● Supported
Flint	`flint`	● Supported
Typhon	`typhoncip30`	● Supported
GeroWallet	`gerowallet`	● Supported

Wallet	CIP-30 Key	Status
Yoroi	`yoroi`	Supported
Vespr	`vespr`	Supported
Begin	`begin`	Supported
NuFi	`nufi`	Supported

Any additional CIP-30 compliant wallet is automatically detected via `window.cardano` scanning.

Security Model

WordPress Nonce Verification

All AJAX endpoints are protected by WordPress nonce verification (`check_ajax_referer`) to prevent CSRF attacks. Nonces are generated server-side and passed to the frontend via `wp_localize_script`.

On-Chain Verification

Payments are verified directly on the Cardano blockchain via Blockfrost API. The plugin checks that the correct amount of ADA was sent to the merchant's address by inspecting transaction UTXOs, preventing spoofed or insufficient payments.

Client-Side Key Isolation

Private keys never leave the user's wallet. The plugin uses the CIP-30 `signTx` API which delegates signing to the wallet extension. The merchant server never has access to customer wallet credentials.

Input Sanitization

All user inputs (TX hashes, order IDs, amounts) are sanitized using WordPress core functions (`sanitize_text_field`, `absint`) and validated with strict regex patterns (64-character hex for TX hashes).

Order Status Protection

Order confirmation AJAX validates that the order belongs to the BeginPay payment method and is in "on-hold" status before allowing status transitions. Already-processed orders cannot be re-confirmed.

Admin Configuration

Setting	Description	Default
Title	Payment method name displayed at checkout	Cardano (ADA)
Description	Custom description shown to customers	Pay with Cardano (ADA)
Cardano Address	Merchant receiving address (bech32 format)	—
CoinMarketCap API Key	API key for real-time ADA/fiat exchange rates	—
Blockfrost API Key	Project ID for on-chain transaction verification	—
QR Code URL	URL to a QR code image of the merchant address	—
TX ID Field	Show/hide the manual transaction ID input field	Enabled
Disclaimer	Custom disclaimer text displayed at checkout	—

Testing & Validation

Test Scenarios Completed

- **Plugin Activation**
  Plugin activates without PHP 8.2 deprecation warnings, all files load correctly, WooCommerce gateway registered
- **Admin Settings**

All gateway settings (API keys, address, QR code, TX ID toggle, disclaimer) save and load correctly

- **Checkout UI Rendering**
  Three-tab payment panel renders correctly with fiat total, ADA conversion, exchange rate, and all interactive elements

- **ADA Rate Conversion**
  Real-time rate fetched from CoinMarketCap API, cart total correctly converted to ADA, refresh button updates rate

- **Wallet Detection & Connection**
  CIP-30 wallet extensions detected via window.cardano, enable() API called successfully, connected address displayed

- **Transaction Building & Signing**
  CBOR transaction built from wallet UTXOs, signed via signTx with partialSign=true, submitted via submitTx

- **On-Chain Verification**
  Blockfrost API confirms transaction with correct amount at merchant address, order status auto-updated to "processing"

- **Copy & QR Code Methods**
  Copy-to-clipboard works for address and amount, QR code displays correctly, tab switching functions properly

- **Order Metadata**
  TX ID, ADA amount, exchange rate, payment method, and verification status correctly saved and displayed in admin, thank-you page, and emails

- **End-to-End Payment**
  Complete flow validated: add to cart → checkout → select ADA → wallet payment → on-chain confirmation → auto-redirect to thank-you page

Technical Highlights

Lightweight CBOR Builder vs CSL

Aspect	CSL (ASM.js)	BeginPay CBOR Builder
File Size	~27 MB	~10 KB
Load Time	~3-5 seconds	<10 ms
Dependencies	WASM or ASM.js runtime	None (vanilla JS)
Output Compatibility	Reference implementation	Byte-identical to CSL v11.5.0
Browser Support	WASM-capable browsers	All browsers (ES5 compatible)
Transaction Types	All Cardano transaction types	Simple ADA transfers

WooCommerce Compatibility

Feature	Status
PHP 8.2+ (no dynamic properties)	Compatible
WooCommerce HPOS (High-Performance Order Storage)	Compatible
Classic Checkout (shortcode-based)	Compatible
WordPress 6.x	Compatible
WooCommerce 7.0+	Compatible

Milestone Deliverables

Requirement	Status	Implementation
WooCommerce payment gateway plugin for	Complete	class-wc-payment-gateway-beginpay.php extending

Requirement	Status	Implementation
Cardano (ADA)		WC_Payment_Gateway
CIP-30 wallet connection and transaction signing	Complete	beginpay-wallet.js with 11 wallet support, signTx/submitTx flow
Real-time ADA/fiat price conversion	Complete	CoinMarketCap API integration with rate caching and AJAX refresh
QR code and manual payment fallback	Complete	Three-tab UI: Connect Wallet, QR Code, Copy Address
On-chain payment verification	Complete	Blockfrost API polling with amount verification at merchant address
Order management with payment metadata	Complete	HPOS-compatible meta: TX ID, ADA amount, exchange rate, verification status
Admin configuration panel	Complete	WooCommerce settings: API keys, address, QR code, disclaimer, TX ID toggle
Modern responsive checkout UI	Complete	Card-based layout with animated status indicators and tab navigation

API Integrations

Service	Purpose	Endpoints Used
CoinMarketCap	Real-time ADA/fiat exchange rate	`GET /v1/cryptocurrency/quotes/latest?symbol=ADA&convert={currency}`
Blockfrost	On-chain transaction verification	`GET /txs/{hash}`, `GET /txs/{hash}/utxos`, `GET /blocks/latest`

Service	Purpose	Endpoints Used
CIP-30	Wallet connection and TX signing	`enable()`, `getUtxos()`, `getChangeAddress()`, `signTx()`, `submitTx()`